

Advanced Digital Design 130323 ModelSim Introduction (Assignment 2) Date: Tuesday, October 24, 2023

Submission:	By not more than 2 people together
Deadline:	Tuesday, November 7, 2023 (17:15) – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

A. Scope:

The purpose of this exercise is to introduce you to Mentor Graphics' **ModelSim**, our tool of choice for logic simulation. We will walk through a simple simulation of a simple inverting gate.

B. Preparations:

- Create a directory for running the lab. The path and directory name must obey to the following requirements:
 - You must have full access “read and write” to that directory.
 - You have enough quota or disk-space to work with that directory.
 - No spaces in the path name and directory name (i.e., “My Documents” is not good).
 - Path name and directory name shall not contain Hebrew characters.
- This exercise contains 3 files: **nnn.vhd**, **nnn.do** and **experiment.vhd**.
Most of this exercise is performed with **nnn.vhd** and **nnn.do**, copy these files to the new directory.
Experiment.vhd is only used in section F (Experiment)

C. You will learn about the following ModelSim commands:

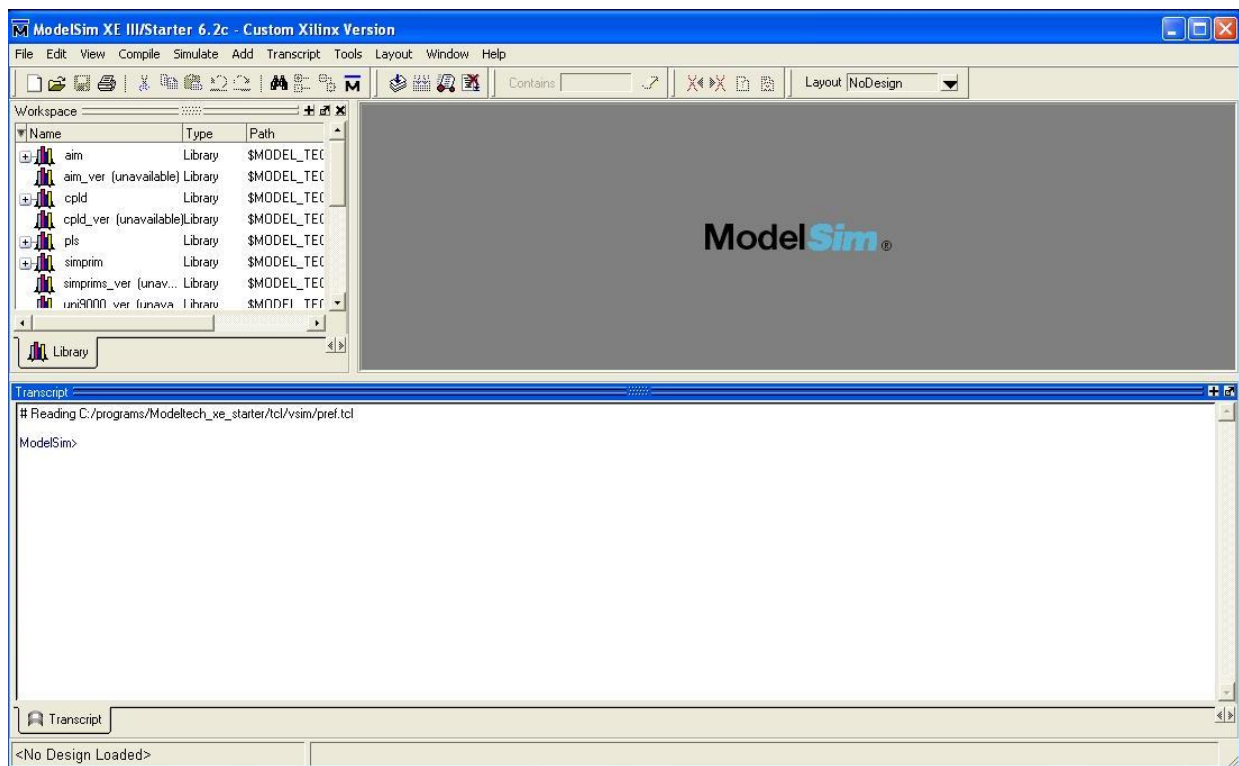
Command	Syntax (simplified)	Description	Example
vlib	vlib library_name	Create a simulation library for HDL objects. The library will be used solely by ModelSim.	vlib work
vcom	vcom file_name.vhd	Compile (analyze) a VHDL file	vcom nnn.vhd
vsim	vsim top_level_object_name	Start simulation for the HDL model in which top_level_object_name is it's top level entity or VHDL configuration.	vsim nnn
	In ModelSim versions to 10.7: The -novopt option for vcom or vsim	Prevent ModelSim from optimizing the design. The simulation might run slowly, but you may get visibility of signals that otherwise will not be seen in waveforms due to optimization	vcom -novopt nnn.vhd vsim -novopt nnn
	From ModelSim version 10.7 and newer: The -voptargs=+acc=lprn option for vsim		vsim -voptargs=+acc=lprn nnn
examine	examine signal_name	Tells ModelSim to print the value of the signal in the transcript window	examine y
run	run duration time_unit	Run simulation for the specified time.	run 100 ns
restart	restart [-f]	Roll back to time 0.	restart -f
force	force signal_name value	Force a value to a signal	force a 0
do	do simulation_script	Run ModelSim commands from a script.	do run.do
quit	quit [-f]	Exit from Modelsim simulator	quit

D. Step by Step Tutorial:

1. Launch ModelSim simulator by double clicking on the ModelSim icon (see figure 1) or by starting it from Windows' "Start Menu"

*Figure 1: ModelSim's Icon*

2. ModelSim simulator is supposed to start, you should see a window similar to the window shown in figure

*Figure 2: ModelSim GUI (Graphical User Interface)*

The GUI (Graphical User Interface) contains several windows. The name of each window (Workspace, Transcript, Objects and alike) is usually given at the top left corner of the window and at the bottom left tab of that window.

3. Tell ModelSim to change to the directory that you created according to the guidelines in **B** above. This should be done by the “File → Change Directory” menu, as shown in figure 3:

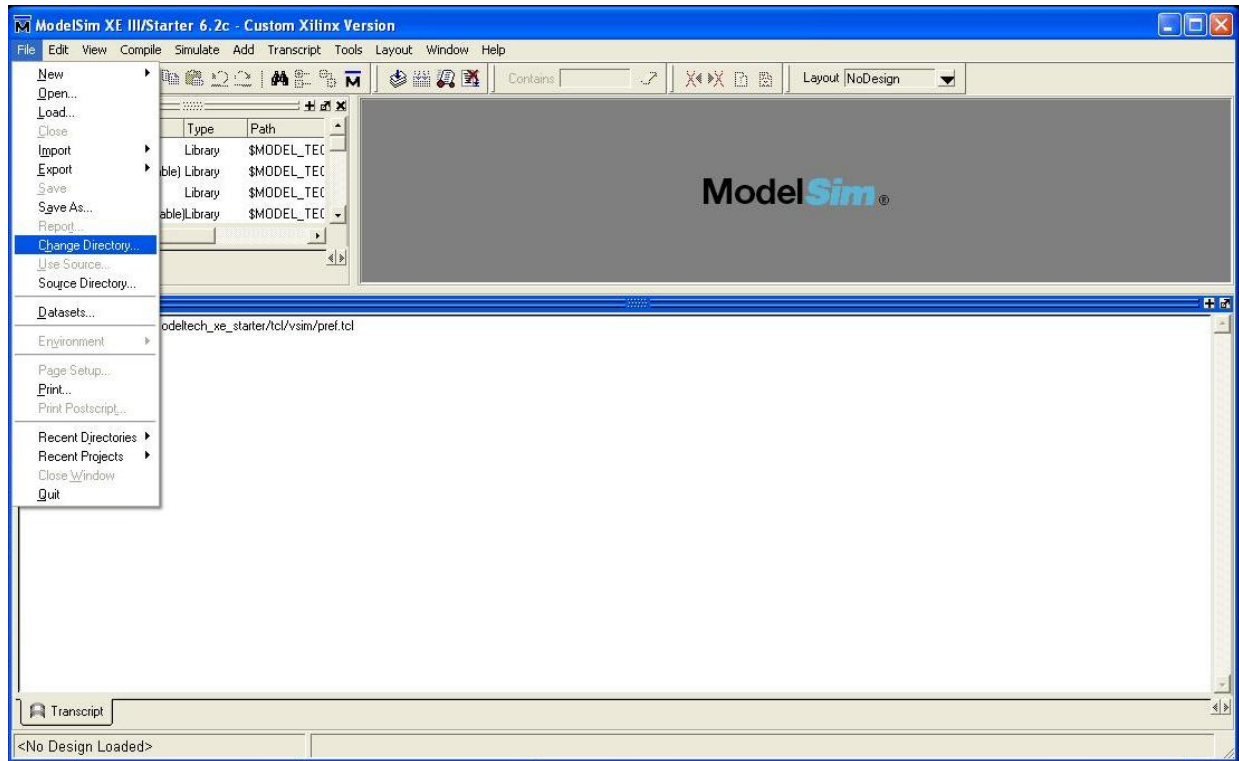


Figure 3: File -> Change Directory Menu

4. Prepare a VHDL work library by using the following command in the transcript window:
5. Prepare (Compile) the VHDL file for simulation by using the following command in the transcript window:
6. Start simulation by using the following command in the transcript window:

```
vlib work
```

```
vcom nnn.vhd
```

```
vsim -voptargs=+acc=lprn nnn
```

Select the signals of nnn from the Objects (or Signals) window and send them to the Waveform window by marking the signals and opening a menu by clicking the right mouse button. In the menu, select “Add to Wave → Selected Signals” option. See Figure 4 for more details.

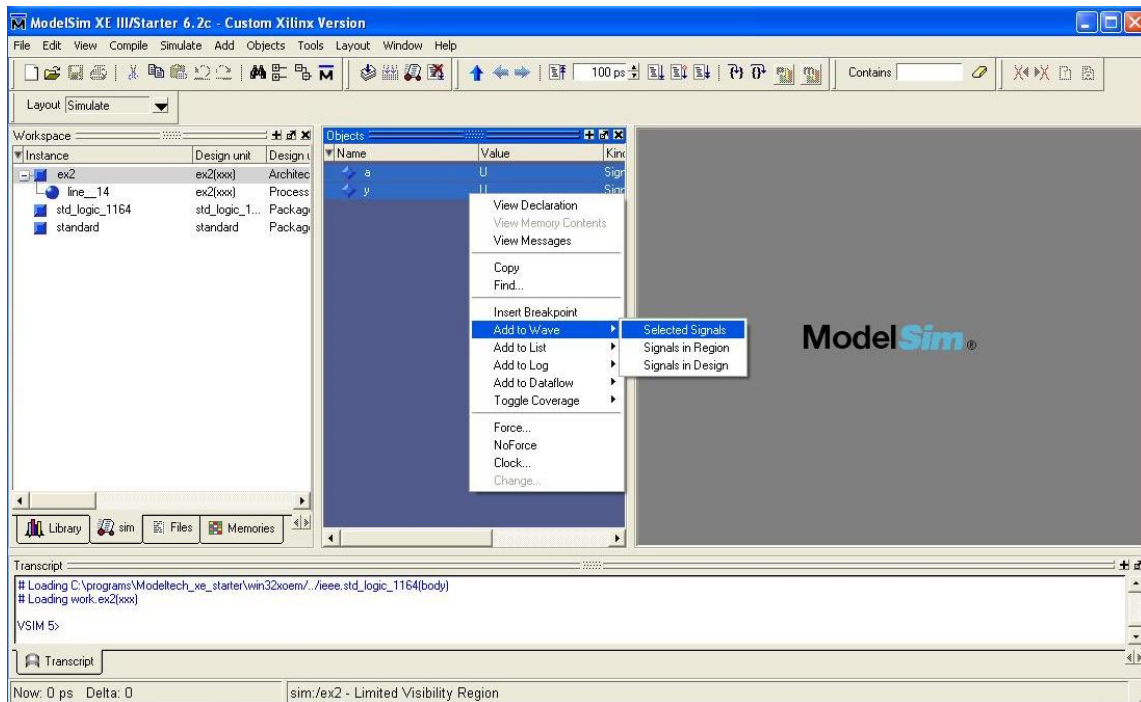


Figure 4: Selecting Signals and Sending them to the Wave Window

You will see a new Waveform window in ModelSim, as illustrated in figure 5:

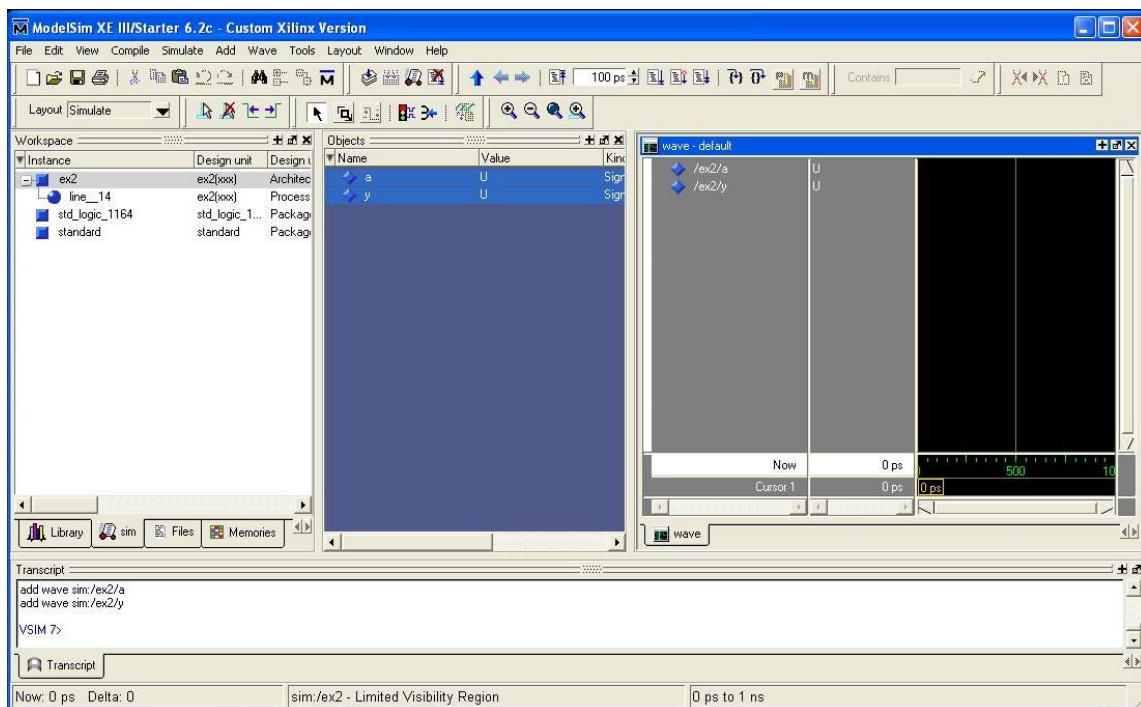


Figure 5: With Waveform Window

7. Start running the following set of commands (as you already noticed, all shall be given in the transcript window) of forcing a value to the design's input signal, running ModelSim for a given period of time changing the value of the signal and running ModelSim for another period of time:

```

force a 0
run 50 ns
force a 1
run 100 ns
force a 0
run 350 ns

```

8. We shall now investigate the Waveform window. The waveform Window is tiled inside ModelSim's GUI, and is now only showing the end of the simulation, as in figure 6:

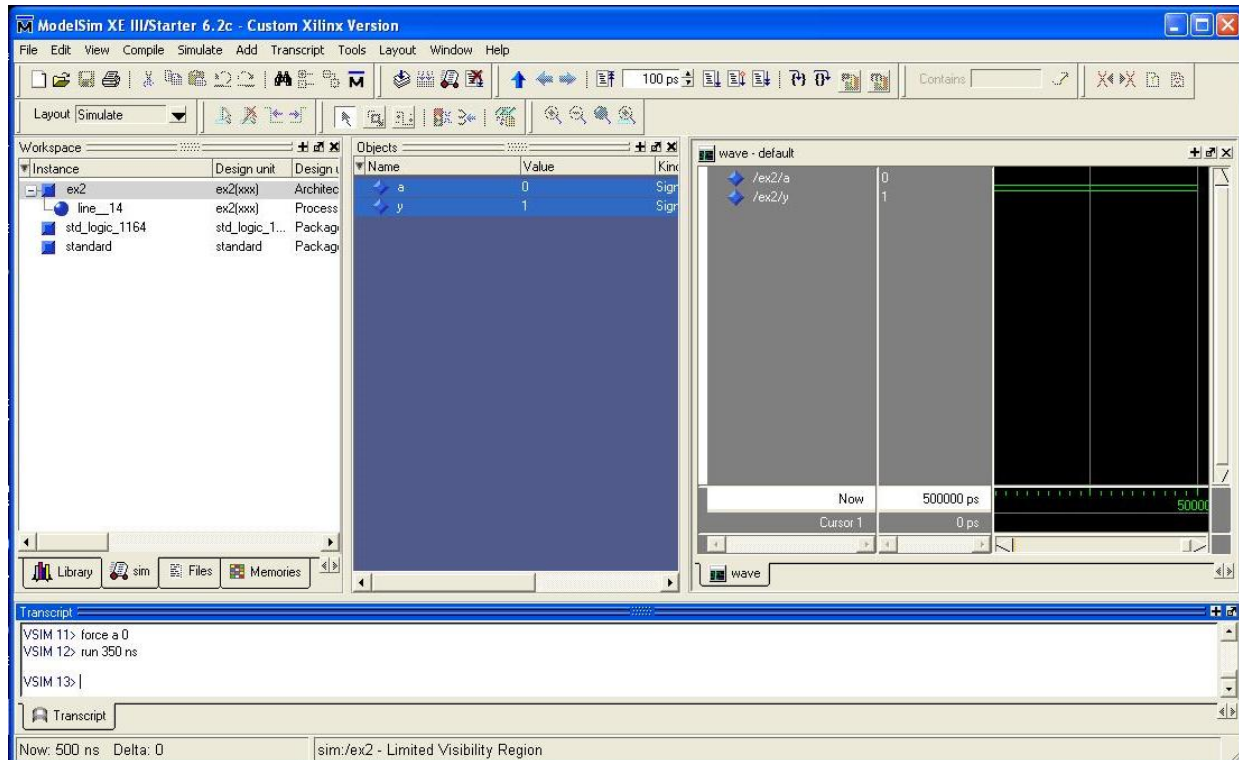


Figure 6: Waveform Window Docked Inside ModelSim's GUI

To carefully examine the waveforms (i.e., the results of the simulation !!!) you need to “undock” the waveform window from the GUI. You should “undock” the window with a left mouse click on the “nail” icon at the top right corner of that window. The waveform window will now be a window of its own, outside of the main ModelSim GUI, as shown in figure 7:

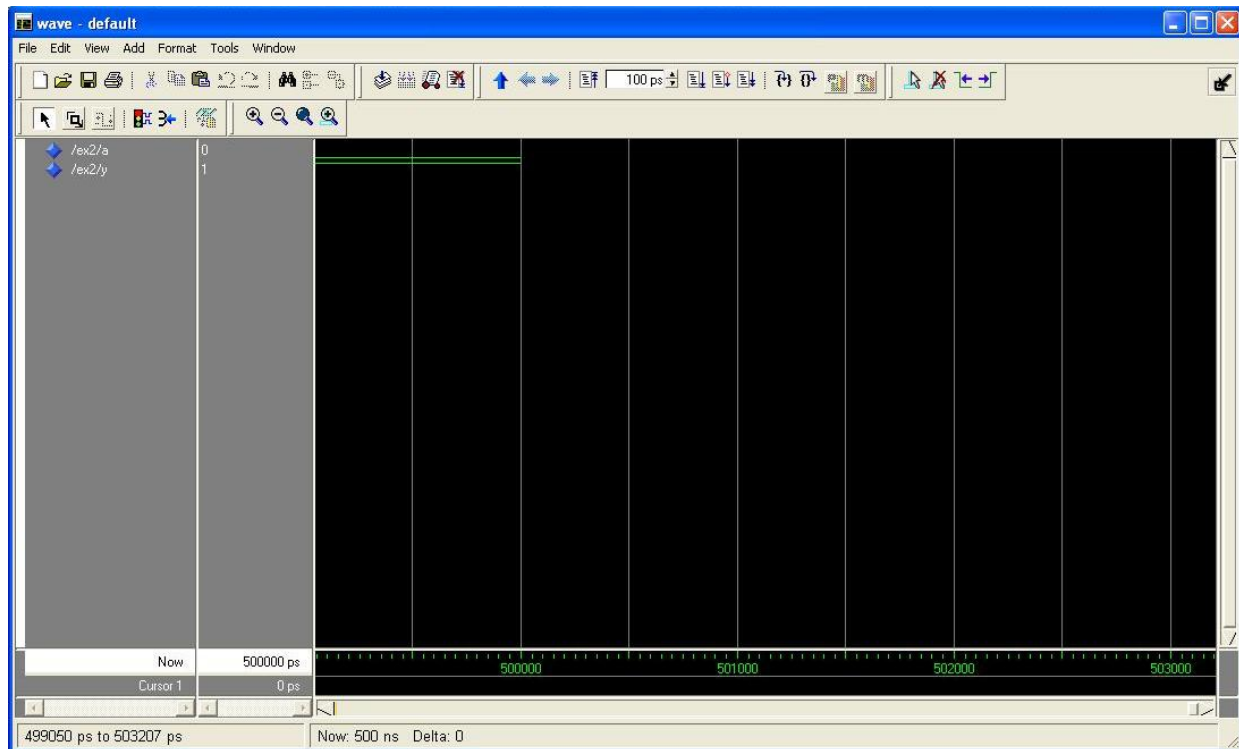


Figure 7: Waveform Window Undocked

Perform a left mouse click on the magnifying glasses icons at the icon's bar of the Waveform window. You should select the “full zoom” magnifying glass to see the whole range of simulation, as shown in figure 8:

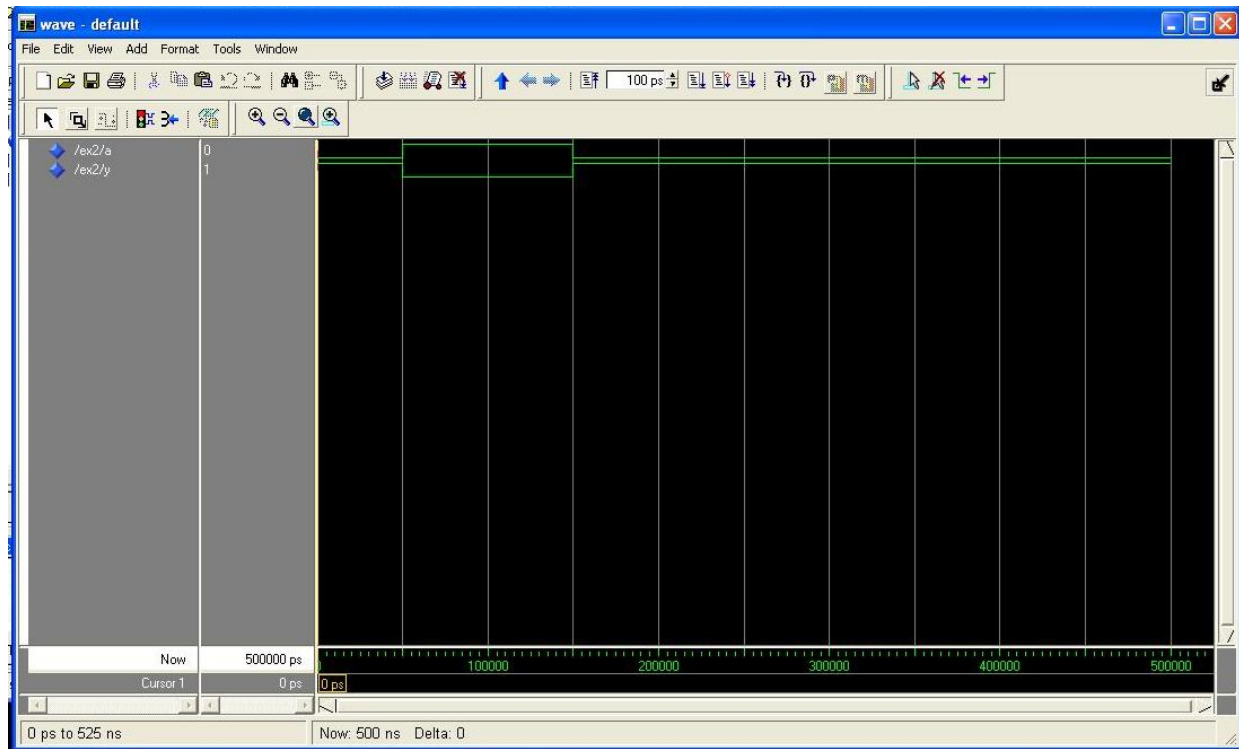


Figure 8: Full Zoom of the Waveform Window

9. Now, roll the simulation back to time zero by typing the following command in the Transcript window:

```
restart -f
```

You will see that the Waveform window will be empty.

10. Play a bit with the force and run commands sequence by forcing a value on 1 and then run ModelSim for a requested period.

Let's add another command, the *examine* command, it will show the value of a selected signal on the transcript window:

```
force a 0
run 50 ns
examine y
force a 1
run 100 ns
examine y
force a 0
run 350 ns
examine y
```

11. Now run a predefined sequence of ModelSim commands by using using nnn.do file from the transcript window, as follows:

```
do nnn.do
```

Full zoom the waveform window and make sure that you understand what and how each line of nnn.do effected the simulation.

12. Exit ModelSim by typing the following command in the transcript window:

```
quit -f
```

E. Playing Around with the “do” File:

Change the “nnn.do” file in such that it would toggle nnn's input every 50 ns (instead of every 100 ns). Run the new nnn.do in ModelSim. You must include the modified “nnn.do” and the resulting waveforms in your report.

F. Experiment (experiment.vhd):

Close ModelSim, create a **new, different** directory, and copy **experiment.vhd** into it.

1. Look into experiment.vhd and write in your report what is the entity name in this file.
2. Is the entity name consistent with the file name? Explain!
3. Open ModelSim again, change directory to the new directory, and run the following commands:

```
vlib work
vcom experiment.vhd
```

4. Now, try each of the following commands and write (in your lab report...) what happens when you perform each one of them. **Some of these commands generate errors and fail... so, write down all your findings in the lab report.**

- a. `vsim -voptargs=+acc=lprn experiment.vhd`
- b. `vsim -voptargs=+acc=lprn experiment`
- c. `vsim -voptargs=+acc=lprn nnn`

When you have something without errors, run a few “force” and “run” commands to see that things are working right (hint: one of the above commands will not give errors, but will **not** respect the “force” and the “run” commands) ...

5. As a conclusion:
 - i. What kind of argument must be given to vsim?
 - ii. What kind of argument must be given to vcom?

G. Your report must include the followings items and answers:

1. Spec for nnn.vhd: What is “nnn.vhd”? What are the input signals? What are the output signals? What does the “nnn” design supposed to do?
2. Write a short and simple ATP (Acceptance Test Plan) for “nnn.vhd”.
3. What should the “nnn.do” script do?
4. Your report must include the waveform of **D.11**, and the transcript window that reflects the examine commands
5. **Answer in your report: Why should we want to use the “examine” command instead (or in addition) to a waveform?**
6. Your report must include the modified “nnn.do” and the resulting waveforms of step E.
7. Your report must include explicit answers to **F.1**, **F.2** and **F.5**
8. Your report must include the observations report, required in section F.